US Patent & Trademark Office
Patent Public Search | Text View

United States Patent	12412791
Kind Code	B2
Date of Patent	September 09, 2025
Inventor(s)	Tsai; Po-Cheng

Liquid-tight structure with a fastener component

Abstract

A liquid-tight structure includes a stepped fastener having a first shank and a second shank. The stepped fastener is threadingly engaged with an opening that includes a first step and a second step. A first diameter of the first shank is larger than a second diameter of the second shank. The liquid-tight structure includes a first seal seated on and in physical communication with a first top surface of the first step of the opening. The first seal is compressed between a first side of the opening and the first shank. The liquid-tight structure includes a second seal seated on and in physical communication with a second top surface of the second step of the opening. The second seal is compressed between a second side of the opening and the second shank.

Inventors:	**Tsai; Po-Cheng (New Taipei, TW)**
Applicant:	**DELL PRODUCTS L.P.** (Round Rock, TX)
Family ID:	**91761229**
Assignee:	**Dell Products L.P. (Round Rock, TX)**
Appl. No.:	**18/151833**
Filed:	**January 09, 2023**

Prior Publication Data

Document Identifier	Publication Date
US 20240237281 A1	Jul. 11, 2024

Publication Classification

Int. Cl.: **H01L23/04** (20060101); **F16J15/02** (20060101); **F16L23/00** (20060101); **H01L23/40** (20060101); H05K7/20 (20060101)

U.S. Cl.:

CPC **H01L23/04** (20130101); **F16J15/021** (20130101); **F16L23/00** (20130101); **H01L23/4006** (20130101); H05K7/20272 (20130101)

Field of Classification Search

USPC: None

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
7527300	12/2008	Li	285/332	F16L 33/223
9699938	12/2016	Shelnutt et al.	N/A	N/A
10168110	12/2018	Krug, Jr.	N/A	B21D 53/02
10863652	12/2019	Conroy	N/A	H05K 7/20763
11092269	12/2020	Kujawski, Jr.	N/A	F16L 21/06
12111000	12/2023	Nowicki	N/A	F16L 37/16
2013/0312854	12/2012	Eriksen	137/544	F16L 37/32
2020/0049429	12/2019	Koontz	N/A	H01L 23/4006
2020/0378858	12/2019	Curtis et al.	N/A	N/A
2022/0095481	12/2021	Ma et al.	N/A	N/A
2024/0039186	12/2023	Su	N/A	H01R 4/70

Primary Examiner: Smith; Courtney L

Attorney, Agent or Firm: Larson Newman, LLP

Background/Summary

FIELD OF THE DISCLOSURE

(1) The present disclosure generally relates to information handling systems, and more particularly relates to a liquid-tight structure with a fastener component for an information handling system.

BACKGROUND

(2) As the value and use of information continues to increase, individuals and businesses seek additional ways to process and store information. One option is an information handling system. An information handling system generally processes, compiles, stores, or communicates information or data for business, personal, or other purposes. Technology and information handling needs and requirements can vary between different applications. Thus, information handling systems can also vary regarding what information is handled, how the information is handled, how much information is processed, stored, or communicated, and how quickly and efficiently the information can be processed, stored, or communicated. The variations in information handling systems allow information handling systems to be general or configured for a specific user or specific use such as financial transaction processing, airline reservations, enterprise data storage, or global communications. In addition, information handling systems can include a variety of hardware and software resources that can be configured to process, store, and communicate information and can include one or more computer systems, graphics interface systems, data storage systems, networking systems, and mobile communication systems. Information handling

systems can also implement various virtualized architectures. Data and voice communications among information handling systems may be via networks that are wired, wireless, or some combination.

SUMMARY

(3) A liquid-tight structure has a stepped fastener that includes a first shank and a second shank. The stepped fastener is threadingly engaged with an opening that includes a first step and a second step. The first diameter of the first shank may be larger than a second diameter of the second shank. The liquid-tight structure includes a first seal seated on and in physical communication with a first top surface of the first step of the opening. The first seal is compressed between a first side of the opening and the first shank. The liquid-tight structure includes a second seal seated on and in physical communication with a second top surface of the second step of the opening. The second seal is compressed between a second side of the opening and the second shank.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) It will be appreciated that for simplicity and clarity of illustration, elements illustrated in the Figures are not necessarily drawn to scale. For example, the dimensions of some elements may be exaggerated relative to other elements. Embodiments incorporating teachings of the present disclosure are shown and described with respect to the drawings herein, in which:

(2) FIG. **1** is a perspective view illustrating a liquid-based cooler according to an embodiment of the present disclosure;

(3) FIG. **2** is a perspective view illustrating a coolant reservoir which is a liquid-tight structure with a fastener component, according to an embodiment of the present disclosure

(4) FIG. **3** is a perspective view illustrating a portion of a liquid-tight structure with a fastener component, according to an embodiment of the present disclosure;

(5) FIG. **4** is a cross-sectional view of a portion of a coolant reservoir taken along line A-A in FIG. **2**, according to an embodiment of the present disclosure; and

(6) FIG. **5** is a block diagram illustrating an information handling system according to an embodiment of the present disclosure.

(7) The use of the same reference symbols in different drawings indicates similar or identical items.

DETAILED DESCRIPTION OF THE DRAWINGS

(8) The following description in combination with the Figures is provided to assist in understanding the teachings disclosed herein. The description is focused on specific implementations and embodiments of the teachings and is provided to assist in describing the teachings. This focus should not be interpreted as a limitation on the scope or applicability of the teachings.

(9) As processors, graphics cards, random access memory, and other components of an information handling system have increased in clock speed and power consumption, the amount of heat produced by such components during normal operation has also increased. Often, the temperatures of these components need to be kept within a reasonable range to prevent overheating, instability, malfunction, and damage leading to a shortened component lifespan. Accordingly, liquid-based coolers, such as a liquid air assistant cooling module or similar have been used to accelerate heat dissipation. However, one disadvantage to using liquid-based coolers is that a coolant reservoir may leak over time. As such, a liquid-tight coolant reservoir is disclosed herein to prevent or eliminate the risk of fluid leaking from a liquid-based cooler.

(10) FIG. **1** shows a liquid-based cooler **100** for cooling an information handling system, such as information handling system **500** of FIG. **5**. Liquid-based cooler **100**, also referred to as a liquid cooling system, includes a radiator **105**, a coolant reservoir **110**, a tube **115**, a pump **120**, a tube **125**, a pump **130**, and a coolant reservoir **140**. Pumps **120** and **130** may be coupled to tubes **115** and

125 and may include any mechanical or electro-mechanical system, apparatus, or device operable to produce a flow of fluid. For example, pumps **120** and **130** may provide fluid flow by applying pressure to coolant in tubes **115** and **125**. The functions and features of a liquid-based cooler **100** are known in the art and will not be further disclosed herein, except as needed to illustrate the various embodiments disclosed herein. The components shown are not drawn to scale and liquid-based cooler **100** may include additional or fewer components. In addition, connections between components may be omitted for descriptive clarity.

(11) Radiator **105** may include any device, system, or apparatus configured to transfer thermal energy from one medium, such as coolant in coolant reservoir **110**, to another for cooling and heating. In some embodiments, radiator **105** may include channels and/or conduits, such as tubes **115** and **125**, and pumps **120** and **130**. Radiator **105** includes coolant reservoirs **110** and **140** that are filled with coolant fluid. The coolant fluid, such as water, betaine, deionized water, or similar, is driven by pumps **120** and **130** to pass through a cold plate via tubes **115** and **125** and back to the radiator to cool down.

(12) Coolant reservoir **110** may include a coolant inlet to fill the reservoir with coolant fluid. Coolant reservoir **110** is a liquid-tight or watertight structure with a coolant inlet that uses a fastener **135** to prevent the coolant fluid from leaking out. Fastener **135** may be a stepped fastener, such as a stepped screw that may be used with a seal such as an “O-ring.” The seal may be constructed of any suitable material and formed in any suitable shape. The seal may be used to prevent the coolant fluid from leaking out around fastener **135** as liquid leaks within the information handling system may cause damage. For example, the fluid leak may cause corrosion to components of the information handling system and or damage to the electrical or electronic circuitry of the information handling system. However, the quality of the tolerance of the O-ring is hard to control and is typically imprecise due to inconsistent manufacturing processes. This affects the liquid-tight compression value of the O-ring making the liquid-tight seal of the screw unreliable. Accordingly, to address this and other concerns, an improved liquid-tight structure with a fastener and seal combination is disclosed herein. Coolant reservoir **140** is similar to coolant reservoir **110**. Accordingly, coolant reservoir **140** includes a fastener **145** which is similar to fastener **135**. Thus, although the descriptions herein are disclosed using coolant reservoir **110** and fastener **135**, such descriptions are also applicable to coolant reservoir **140** and fastener **145**.

(13) Those of ordinary skill in the art will appreciate that the configuration of the liquid-based cooler depicted in FIG. **1** and coolant reservoir **110** in FIG. **2** may vary. For example, the illustrative components within liquid-based cooler **100** and coolant reservoir **110** are not intended to be exhaustive, but rather are representative to highlight components that can be utilized to implement aspects of the present disclosure. For example, other devices and/or components may be used in addition to or in place of the devices/components depicted. The depicted example does not convey or imply any architectural or other limitations with respect to the presently described embodiments and/or the general disclosure. For example, although coolant reservoir **110** is shown to be rectangular, those of ordinary skill in the art will appreciate that coolant reservoir **110** is not limited to such a shape. Accordingly, coolant reservoir **110** may be square, round, cylindrical, etc. Further, embodiments of the present disclosure do not only apply to coolant reservoirs but to any liquid reservoir wherein providing leak-proof functionality is needed or desired. In addition, although coolant reservoir **110** is shown as part of a liquid-based cooler, the leak-proof structure may be included in other systems wherein liquid contained in a leak-proof structure is needed or desired. In the discussion of the figures, reference may also be made to components illustrated in other figures for the continuity of the description.

(14) FIGS. **2** and **3** show the coolant reservoir **110** which includes a fastener component and seal components. Coolant reservoir **110** is a liquid-tight structure which is also referred to herein as a leak-proof structure or a watertight structure. Coolant reservoir **110** may be included in liquid-based cooler **100** of FIG. **100**. Coolant reservoir **110** includes fastener **135**, a seal **305**, a seal **310**,

and an opening **315** with multiple stepped portions and a bottom. In this example, opening **315** includes step **320**, step **325**, and step **340**. Fastener **135** includes a head **330** and a threaded shank **335**. To form a liquid-tight structure, fastener **135** may be configured to tightly fit in opening **315** with seals **305** and **310**, such that the top of fastener **135** is flushed with the top of coolant reservoir **110** and the bottom of threaded shank **335** may be flushed with the bottom of opening **315**. Seals **305** and **310** may be compressed against the wall of opening **315** by tightening fastener **135** into the inner surface of coolant reservoir **110** via opening **315** further sealing the gap between the fastener head and the surface and/or interior sides of coolant reservoir **110**. Thus, achieving liquid-tight functionality.

(15) Fastener **135** may be a screw, a thumb screw, a step screw, a shoulder screw, a bolt, or any type of connecting member. Fastener **135** may be made of suitable material, such as various metals that include stainless steel, brass, titanium, etc. Fastener **135** may also be made from a plastic material with suitable hardness or strength. In one embodiment, fastener **135** may be a stepped screw that is secured through a screw hole, such as opening **315**. Fastener **135** may be a threaded fastener that includes a head **330** and threaded shank **335**. Head **330** may be configured such that the top portion may be flushed to the top surface of coolant reservoir **110** when secured through opening **315**. Further, head **330** is of a height, wherein the bottom portion of head **330** may be flushed to a grove of opening **315**, such as step **320** when tightened. Because of the rigid contact between the head **330** of fastener **135** and opening **315**, seals **305** and **310** may not be over-compressed when threaded shank **335** is screwed through opening **315**.

(16) Seals **305** and **310** may be configured to provide a liquid-tight feature to fastener **135** preventing coolant from leaking from opening **315**. The liquid-tight feature may be a result of a frictional fit between the seals **305** and **310**, fastener **135**, and/or sides of opening **315**. Seals **305** and **310** may be formed with suitable material such as silicone or rubber using a process such as injection molding. In one embodiment, seals **305** and **310** may include O-rings or gaskets of similar shape. Seals **305** and **310** may have a suitable thickness to provide a fluid seal with fastener **135** preventing a fluid or liquid leak. Seal **305** may be configured such that seal **305** is disposed at a groove of opening **315**, such as in physical communication with step **320** when fastener **135** is screwed through opening **315**. Similarly, seal **310** may be configured such that seal **310** is disposed at a groove of opening **315**, such as in physical communication with step **340** when fastener **135** is screwed through opening **315**.

(17) Seal **305** may be configured as an exterior seal with a higher hardness than seal **310** which is an interior seal as it is disposed below seal **305**. The different hardness of seals **305** and **310** may help avoid the risk of monotonous tolerance. The durometer, also referred to as shore hardness, typically runs numerically from 0-100 with a lower number indicating a softer material and a higher number indicating a harder material. As such, seal **305** may have a higher durometer than seal **310**. Accordingly, seal **310** may be a fraction of the durometer of seal **305**. In one embodiment, seal **310** may have half the durometer number of seal **305**. For example, seal **310** may have a 60A durometer while seal **305** may have a 30A durometer. In another embodiment, seal **310** may have more or less than half the durometer number of seal **305**. For example, the durometer value of seal **310** may be a quarter or three-quarters of the value of the durometer of seal **305**. The combination of seals **305** and **310** may be configured to fill up space between opening **315** and fastener **135** forming a leak-proof seal and precluding void at a sealing interface.

(18) FIG. **4** shows a cross-sectional view of coolant reservoir **110** taken along line A-A as shown in FIG. **2**. In this example, coolant reservoir **110** includes fastener **135** that is threadedly engaged and in physical communication with opening **315**, wherein fastener **135** may have reached or is about to reach the bottom of opening **315**. Fastener **135** includes head **330** which has a diameter that fits in opening **315** such that an outer surface of head **330** is disposed to and in physical communication with side **415** of opening **315**, wherein side **415** is the top side of the stepped opening and is disposed or abutted to step **320**. Further, a top surface of head **330** is substantially flush with a top

surface of coolant reservoir **110** while the bottom portion of head **330** may be seated on and in physical communication with step **320** when fastener **135** is fully threadedly engaged to opening **315**.

(19) Fastener **135** also includes an unthreaded shank **405** and threaded shank **335**. Unthreaded shank **405** is disposed between head **330** and threaded shank **335**, wherein head **330** is above unthreaded shank **405**, and wherein threaded shank **335** is below unthreaded shank **405**. In one embodiment, a diameter of unthreaded shank **405** is larger than a diameter of threaded shank **335**. Also, the diameter of head **330** is larger than the diameter of unthreaded shank **405** and the diameter of threaded shank **335**.

(20) Seal **305** may be seated on and in physical communication with step **325** and disposed between unthreaded shank **405** of fastener **135** and side **420** of opening **315** which is disposed to step **325**. In particular, seal **305** may be compressed between a side surface of unthreaded shank **405** and side **420** that is abutted to step **325**. In addition, seal **305** may also be compressed from the top by the bottom surface of head **330**. Seal **305** being compressed from the side, top, and/or bottom may form a leak-tight or liquid-tight seal.

(21) Seal **310** may be disposed between and in physical communication with one side surface of unthreaded shank **410** and a surface of side **425** which is disposed to step **340**. In particular, seal **310** may be compressed between another side surface of unthreaded shank **410** and side **425** that is abutted to step **340**. In addition, seal **310** may also be compressed from the top by the bottom surface of unthreaded shank **405**. Seal **310** being compressed from the side, top, and/or bottom may form a leak-tight or liquid-tight seal. In one embodiment, a diameter of seal **305** is larger than a diameter of seal **310**.

(22) The components shown are not drawn to scale and coolant reservoir **110** may include additional or fewer components. In addition, coolant reservoir **110** may not include each of the components shown in FIG. **3** and FIG. **4**. Additionally, or alternatively, coolant reservoir **110** may include various additional components in addition to those that are shown in FIG. **2** and FIG. **3**. Furthermore, some components that are represented as separate components in FIG. **2** and FIG. **3** may in certain embodiments instead are integrated with other components.

(23) FIG. **5** shows an embodiment of an information handling system **500** including the liquid-based cooler **100**. Information handling system **500** further includes processors **502** and **504**, a chipset **510**, a memory **520**, a graphics adapter **530** connected to a video display **534**, a non-volatile RAM (NV-RAM) **540** that includes a basic input and output system/extensible firmware interface (BIOS/EFI) module **542**, a disk controller **550**, a hard disk drive (HDD) **554**, an optical disk drive **556**, a disk emulator **560** connected to a solid-state drive (SSD) **564**, an input/output (I/O) interface **570** connected to an add-on resource **574** and a trusted platform module (TPM) **576**, a network interface **580**, and a baseboard management controller (BMC) **590**. Processor **502** is connected to chipset **510** via processor interface **506**, and processor **504** is connected to the chipset via processor interface **508**. In a particular embodiment, processors **502** and **504** are connected together via a high-capacity coherent fabric, such as a HyperTransport link, a QuickPath Interconnect, or the like. Chipset **510** represents an integrated circuit or group of integrated circuits that manage the data flow between processors **502** and **504** and the other elements of information handling system **500**. In a particular embodiment, chipset **510** represents a pair of integrated circuits, such as a northbridge component and a southbridge component. In another embodiment, some or all of the functions and features of chipset **510** are integrated with one or more of processors **502** and **504**.

(24) Memory **520** is connected to chipset **510** via a memory interface **522**. An example of memory interface **522** includes a double data rate (DDR) memory channel and memory **520** represents one or more DDR dual in-line memory modules (DIMMs). In a particular embodiment, memory interface **522** represents two or more DDR channels. In another embodiment, one or more of processors **502** and **504** include a memory interface that provides a dedicated memory for the processors. A DDR channel and the connected DDR DIMMs can be in accordance with a particular

DDR standard, such as a DDR3 standard, a DDR4 standard, a DDR5 standard, or the like.
(25) Memory **520** may further represent various combinations of memory types, such as dynamic random access memory (DRAM) DIMMs, static random access memory (SRAM) DIMMs, non-volatile DIMMs (NV-DIMMs), storage class memory devices, read-only memory (ROM) devices, or the like. Graphics adapter **530** is connected to chipset **510** via a graphics interface **532** and provides a video display output **536** to a video display **534**. An example of a graphics interface **532** includes a Peripheral Component Interconnect-Express (PCIe) interface and graphics adapter **530** can include a four-lane (×4) PCIe adapter, an eight-lane (×8) PCIe adapter, a 16-lane (×16) PCIe adapter, or another configuration, as needed or desired. In a particular embodiment, graphics adapter **530** is provided down on a system printed circuit board (PCB). Video display output **536** can include a Digital Video Interface (DVI), a High-Definition Multimedia Interface (HDMI), a DisplayPort interface, or the like, and video display **534** can include a monitor, a smart television, an embedded display such as a laptop computer display, or the like.
(26) NV-RAM **540**, disk controller **550**, and I/O interface **570** are connected to chipset **510** via an I/O channel **512**. An example of I/O channel **512** includes one or more point-to-point PCIe links between chipset **510** and each of NV-RAM **540**, disk controller **550**, and I/O interface **570**. Chipset **510** can also include one or more other I/O interfaces, including a PCIe interface, an Industry Standard Architecture (ISA) interface, a Small Computer Serial Interface (SCSI) interface, an Inter-Integrated Circuit (I.sup.2C) interface, a System Packet Interface (SPI), a Universal Serial Bus (USB), another interface, or a combination thereof. NV-RAM **540** includes BIOS/EFI module **542** that stores machine-executable code (BIOS/EFI code) that operates to detect the resources of information handling system **500**, to provide drivers for the resources, to initialize the resources, and to provide common access mechanisms for the resources. The functions and features of BIOS/EFI module **542** will be further described below.
(27) Disk controller **550** includes a disk interface **552** that connects the disc controller to a hard disk drive (HDD) **554**, to an optical disk drive (ODD) **556**, and to disk emulator **560**. An example of disk interface **552** includes an Integrated Drive Electronics (IDE) interface, an Advanced Technology Attachment (ATA) such as a parallel ATA (PATA) interface or a serial ATA (SATA) interface, a SCSI interface, a USB interface, a proprietary interface, or a combination thereof. Disk emulator **560** permits SSD **564** to be connected to information handling system **500** via an external interface **562**. An example of external interface **562** includes a USB interface, an institute of electrical and electronics engineers (IEEE) **1394** (Firewire) interface, a proprietary interface, or a combination thereof. Alternatively, SSD **564** can be disposed within information handling system **500**.
(28) I/O interface **570** includes a peripheral interface **572** that connects the I/O interface to add-on resource **574**, to TPM **576**, and to network interface **580**. Peripheral interface **572** can be the same type of interface as I/O channel **512** or can be a different type of interface. As such, I/O interface **570** extends the capacity of I/O channel **512** when peripheral interface **572** and the I/O channel are of the same type, and the I/O interface translates information from a format suitable to the I/O channel to a format suitable to the peripheral interface **572** when they are of a different type. Add-on resource **574** can include a data storage system, an additional graphics interface, a network interface card (NIC), a sound/video processing card, another add-on resource, or a combination thereof. Add-on resource **574** can be on a main circuit board, on separate circuit board or add-in card disposed within information handling system **500**, a device that is external to the information handling system, or a combination thereof.
(29) Network interface **580** represents a network communication device disposed within information handling system **500**, on a main circuit board of the information handling system, integrated onto another component such as chipset **510**, in another suitable location, or a combination thereof. Network interface **580** includes a network channel **582** that provides an interface to devices that are external to information handling system **500**. In a particular

embodiment, network channel **582** is of a different type than peripheral interface **572**, and network interface **580** translates information from a format suitable to the peripheral channel to a format suitable to external devices.
(30) In a particular embodiment, network interface **580** includes a NIC or host bus adapter (HBA), and an example of network channel **582** includes an InfiniBand channel, a Fibre Channel, a Gigabit Ethernet channel, a proprietary channel architecture, or a combination thereof. In another embodiment, network interface **580** includes a wireless communication interface, and network channel **582** includes a Wi-Fi channel, a near-field communication (NFC) channel, a Bluetooth® or Bluetooth-Low-Energy (BLE) channel, a cellular based interface such as a Global System for Mobile (GSM) interface, a Code-Division Multiple Access (CDMA) interface, a Universal Mobile Telecommunications System (UMTS) interface, a Long-Term Evolution (LTE) interface, or another cellular based interface, or a combination thereof. Network channel **582** can be connected to an external network resource (not illustrated). The network resource can include another information handling system, a data storage system, another network, a grid management system, another suitable resource, or a combination thereof.
(31) BMC **590** is connected to multiple elements of information handling system **500** via one or more management interface **592** to provide out of band monitoring, maintenance, and control of the elements of the information handling system. As such, BMC **590** represents a processing device different from processor **502** and processor **504**, which provides various management functions for information handling system **500**. For example, BMC **590** may be responsible for power management, cooling management, and the like. The term BMC is often used in the context of server systems, while in a consumer-level device a BMC may be referred to as an embedded controller (EC). A BMC included at a data storage system can be referred to as a storage enclosure processor. A BMC included at a chassis of a blade server can be referred to as a chassis management controller and embedded controllers included at the blades of the blade server can be referred to as blade management controllers. Capabilities and functions provided by BMC **590** can vary considerably based on the type of information handling system. BMC **590** can operate in accordance with an Intelligent Platform Management Interface (IPMI). Examples of BMC **590** include an Integrated Dell R Remote Access Controller (iDRAC).
(32) Management interface **592** represents one or more out-of-band communication interfaces between BMC **590** and the elements of information handling system **500**, and can include an Inter-Integrated Circuit (I.sup.2C) bus, a System Management Bus (SMBUS), a Power Management Bus (PMBUS), a Low Pin Count (LPC) interface, a serial bus such as a Universal Serial Bus (USB) or a Serial Peripheral Interface (SPI), a network interface such as an Ethernet interface, a high-speed serial data link such as a PCIe interface, a Network Controller Sideband Interface (NC-SI), or the like. As used herein, out-of-band access refers to operations performed apart from a BIOS/operating system execution environment on information handling system **500**, that is apart from the execution of code by processors **502** and **504** and procedures that are implemented on the information handling system in response to the executed code.
(33) BMC **590** operates to monitor and maintain system firmware, such as code stored in BIOS/EFI module **542**, option ROMs for graphics adapter **530**, disk controller **550**, add-on resource **574**, network interface **580**, or other elements of information handling system **500**, as needed or desired. In particular, BMC **590** includes a network interface **594** that can be connected to a remote management system to receive firmware updates, as needed or desired. Here, BMC **590** receives the firmware updates, stores the updates to a data storage device associated with the BMC, transfers the firmware updates to NV-RAM of the device or system that is the subject of the firmware update, thereby replacing the currently operating firmware associated with the device or system, and reboots information handling system, whereupon the device or system utilizes the updated firmware image.
(34) BMC **590** utilizes various protocols and application programming interfaces (APIs) to direct

and control the processes for monitoring and maintaining the system firmware. An example of a protocol or API for monitoring and maintaining the system firmware includes a graphical user interface (GUI) associated with BMC **590**, an interface defined by the Distributed Management Taskforce (DMTF) (such as a Web Services Management (WSMan) interface, a Management Component Transport Protocol (MCTP) or, a Redfish® interface), various vendor defined interfaces (such as a Dell EMC Remote Access Controller Administrator (RACADM) utility, a Dell EMC OpenManage Enterprise, a Dell EMC OpenManage Server Administrator (OMSS) utility, a Dell EMC OpenManage Storage Services (OMSS) utility, or a Dell EMC OpenManage Deployment Toolkit (DTK) suite), a BIOS setup utility such as invoked by a “F2” boot option, or another protocol or API, as needed or desired.

(35) In a particular embodiment, BMC **590** is included on a main circuit board (such as a baseboard, a motherboard, or any combination thereof) of information handling system **500** or is integrated onto another element of the information handling system such as chipset **510**, or another suitable element, as needed or desired. As such, BMC **590** can be part of an integrated circuit or a chipset within information handling system **500**. An example of BMC **590** includes an iDRAC, or the like. BMC **590** may operate on a separate power plane from other resources in information handling system **500**. Thus, BMC **590** can communicate with the management system via network interface **594** while the resources of information handling system **500** are powered off. Here, information can be sent from the management system to BMC **590** and the information can be stored in a RAM or NV-RAM associated with the BMC. Information stored in the RAM may be lost after power-down of the power plane for BMC **590**, while information stored in the NV-RAM may be saved through a power-down/power-up cycle of the power plane for the BMC.

(36) Information handling system **500** can include additional components and additional busses, not shown for clarity. For example, information handling system **500** can include multiple processor cores, audio devices, and the like. While a particular arrangement of bus technologies and interconnections is illustrated for the purpose of example, one of skill will appreciate that the techniques disclosed herein are applicable to other system architectures. Information handling system **500** can include multiple central processing units (CPUs) and redundant bus controllers. One or more components can be integrated together. Information handling system **500** can include additional buses and bus protocols, for example, I.sup.2C and the like. Additional components of information handling system **500** can include one or more storage devices that can store machine-executable code, one or more communications ports for communicating with external devices, and various input and output (I/O) devices, such as a keyboard, a mouse, and a video display.

(37) For purposes of this disclosure information handling system **500** can include any instrumentality or aggregate of instrumentalities operable to compute, classify, process, transmit, receive, retrieve, originate, switch, store, display, manifest, detect, record, reproduce, handle, or utilize any form of information, intelligence, or data for business, scientific, control, entertainment, or other purposes. For example, information handling system **500** can be a personal computer, a laptop computer, a smartphone, a tablet device or other consumer electronic device, a network server, a network storage device, a switch, a router, or another network communication device, or any other suitable device and may vary in size, shape, performance, functionality, and price. Further, information handling system **500** can include processing resources for executing machine-executable code, such as processor **502**, a programmable logic array (PLA), an embedded device such as a System-on-a-Chip (SoC), or other control logic hardware. Information handling system **500** can also include one or more computer-readable media for storing machine-executable code, such as software or data.

(38) Although the present disclosure has been described in detail, it should be understood that various changes, substitutions, and alterations can be made hereto without departing from the spirit and scope of the disclosure as defined by the appended claims.

Claims

1. A liquid-tight structure comprising: a stepped screw that includes a first shank and a second shank, the stepped screw is screwed in a screw hole that includes a first step and a second step, wherein a first diameter of the first shank is larger than a second diameter of the second shank, wherein the stepped screw further includes a head, and wherein a bottom portion of the head is seated on a third step of the screw hole; a first O-ring seated on and in physical communication with a first top surface of the first step of the screw hole, wherein the first O-ring is compressed between a first side of the screw hole and the first shank forming a first seal; and a second O-ring seated on and in physical communication with a second top surface of the second step of the screw hole, wherein the second O-ring is compressed between a second side of the screw hole and the second shank.
2. The liquid-tight structure of claim 1, wherein the second shank is a threaded shank.
3. The liquid-tight structure of claim 1, wherein the first O-ring and the second O-ring are liquid-tight.
4. The liquid-tight structure of claim 1, wherein the liquid-tight structure is used in a liquid-based cooler.
5. The liquid-tight structure of claim 1, wherein the first side of the screw hole is abutted to the first step.
6. The liquid-tight structure of claim 1, wherein the second side of the screw hole is abutted to the second step.
7. The liquid-tight structure of claim 1, wherein the first O-ring has a higher shore hardness than the second O-ring.
8. The liquid-tight structure of claim 1, wherein diameter of the head is greater than the first shank.
9. The liquid-tight structure of claim 1, wherein a top portion of the head is flushed with a top surface of the liquid-tight structure.
10. An information handling system with a liquid cooling system comprising: a stepped screw that includes a first shank and a second shank, wherein the stepped screw is screwed into a screw hole that includes a first step and a second step, wherein a first diameter of the first shank is larger than a second diameter of the second shank, wherein the stepped screw further includes a head, and wherein a bottom portion of the head is seated on a third step of the screw hole; a first O-ring seated on and in physical communication with a first top surface of the first step of the screw hole, wherein the first O-ring is compressed between a first side of the screw hole and the first shank forming a first seal; and a second O-ring seated on and in physical communication with a second top surface of the second step of the screw hole, wherein the second O-ring is compressed between a second side of the screw hole and the second shank forming a second seal.
11. The information handling system of claim 10, wherein the stepped screw further includes a threaded shank.
12. The information handling system of claim 10, wherein the first side of the screw hole is abutted to the first step.
13. The information handling system of claim 10, wherein the second side of the screw hole is abutted to the second step.
14. The information handling system of claim 10, wherein the first O-ring has a higher shore hardness than the second O-ring.
15. The information handling system of claim 14, wherein the second shank is threaded.
16. A leak-proof structure comprising: a stepped screw that includes a first shank and a second shank, the stepped screw is screwed into a screw hole that includes a first step and a second step, wherein a first diameter of the first shank is larger than a second diameter of the second shank, wherein the stepped screw further includes a head, and wherein a bottom portion of the head is

seated on a third step of the screw hole; a first O-ring seated on and in physical communication with a first top surface of the first step of the screw hole, wherein the first O-ring is compressed between a first side of the screw hole and the first shank forming a first seal; and a second O-ring seated on and in physical communication with a second top surface of the second step of the screw hole, wherein the second O-ring is compressed between a second side of the screw hole and the second shank forming a second seal and wherein the first seal has a higher shore hardness than the second seal.

17. The leak-proof structure of claim 16, wherein the leak-proof structure is used for storing coolant for a liquid-based cooler of an information handling system.

18. The leak-proof structure of claim 16, wherein the first shank is an unthreaded shank.

19. The leak-proof structure of claim 16, wherein the first side of the screw hole is abutted to the first step.

20. The leak-proof structure of claim 16, wherein the second side of the screw hole is abutted to the second step.